

As a **human computer interaction (HCI)** and **social computing** researcher, my work focuses on *understanding and reshaping how disruptive technologies (e.g. AI-mediated labor markets) interact with and affect the present and future of workers*. To promote more collective, safe and prosperous worker communities, I draw on design and qualitative approaches (e.g. multi-stakeholder co-design/interviews in solidarity with workers), quantitative empirical methods (e.g. statistical analysis, LLM-supported interventions) and system development (e.g. interaction prototypes).

One type of worker I study is **platform-based gig workers**, whose labor and services are increasingly woven into the fabric of everyday life. For consumers, gigs offer convenient, on-demand help with everyday tasks – e.g., Uber, caretaking, creative and professional projects – by drawing on the wide-ranging expertise of a global workforce. For workers, the temporary and task-based nature of gigs represents *independence* from traditional jobs and organizations – affording them the perceived freedom to choose not only *when* and *where* to labor¹, but also *what* to work on. However, as platforms mediate between workers and consumers, their hegemonic data practices shift significant agency and autonomy away from both stakeholder groups [a]. To centralize power, platforms capitalize on the absence of legal protections for independent contractors to exploit gig workers physically, financially and psychologically [b]. With rising rates of worker turnover and displacement, how do we create transformative interventions that propel workforces toward more natural, collective and liberated future interactions? Overall, the primary themes of my research:

- ① employs large-scale computational methods to identify worker practices [1] and intentions [11]
- ② co-designs with multi-stakeholder communities to envision collective, sustainable futures [2, 5, 6]
- ③ develops [7] interventions that guide more joyful [10], pluralistic [3, 9] and natural interactions.

At the intersection of the HCI, design and CSCW² communities, my thesis work applied human-centered techniques to understand *interactions between disruptive technologies (i.e. gig platforms) and stakeholders (i.e. workers, consumers, regulators)*, followed by the **design and development of alternative interventions** to probe and reshape oppressive technology-mediated relations. I contributed to 12 publications/workshop (7 as first-author, 2 as advising author) across a range of venues: ACM CHI, CSCW, UIST, DIS, CHIWORK and IEEE VL/HCC. Two of my first-authored publications received best paper and honorable mention recognitions at CHI and CHIWORK, and I acquired over \$224,000 in funding as a NSF GRFP recipient and a Block Center for Technology and Society grant awardee.

By investigating timely challenges through multi-stakeholder and multi-scalar approaches, I aim to continue *surfacing timeless, generalizable designs and interventions* that guide future AI-mediated workforces in navigating disruptive innovations. I am passionate about creatively applying my empirical and design expertise to promote futures of work that (1) evolve individual human capabilities (2) motivate collective and supportive social relations and (3) cultivate deliberative, trusting and pluralistic communities.

① Understanding Worker Practices & Rationales via Large-Scale Data Analysis [1, 11]

Gig platforms commonly leverage a consumer-first rhetoric to justify and avoid accountability for unfair labor practices. But despite highly competitive, isolating platform environments, workers continue to willingly engage – suggesting the presence of underexamined intrinsic motivations that drive them to develop strategies for handling unexpected challenges. In my first pair of contributions, I present large-scale analyses that uncover (1) *how* freelancers communicate on platforms to achieve productivity and success (Fig. 1) as well as (2) crowdworker rationales on *why* they perform platform-mediated gigs despite harsh working conditions.

¹ For example, Uber marketed work flexibility through the slogan: “Drive when you want. Earn what you need.”

² Computer-supported cooperative work

To **scalably understand individual worker strategies**, we engaged in multi-month negotiations with collaborators to access a large-scale dataset (> 2 million messages, >58k freelancers, >56k projects) from a leading online freelancing platform. Applying regression analyses on these platform-mediated freelancer-client communications, I found statistically significant correlations between (1) personalized message content and higher *job acquisition rates* (2) standardized messaging schedules and increased *job completion rates*, as well as (3) repeated template use across projects that associated with greater bid volumes and *long term revenue* [1].

But *why* are workers motivated to continue engaging in online platform labor despite subpar conditions? Most workers' underlying beliefs about platforms likely remained undisclosed, under-examined, and misguided by platform-constructed illusions of flexibility. To explore such latent *grievances* and *rationales*, we leveraged the persuasive power of language models to intervene and dissuade crowdworkers from believing five common myths around gig work conditions. In our sample, 201 of 263 Prolific crowdworkers believed at least one myth, and by eliciting their rationales and responding with persuasive LLM responses, we revealed how **crowdworkers labor out of intrinsic desires to preserve independence, autonomy and flexibility** [11].

② Co-designing Multi-Stakeholder Interventions for Platform Accountability [2, 5, 6, 7]

Mobilizing the data and efforts of online gig workforces revealed crucial strategies and rationales. But offline gig work requires support from higher-power stakeholders to enact infrastructural reforms that advance gig labor rights and protections.

To synergistically integrate priorities from multiple stakeholders, I led co-design workshops with policy experts, platform employees and workers to tackle five representative problems in gig work [2].

Our solution-oriented version of the speed-dating design method led stakeholders to generate and rank infrastructural interventions they are most prepared or motivated to implement (Fig. 2), which include visions of *stronger regulation*, *worker-owned coops* and *platform-government co-regulation*. Stakeholders also envisioned more incremental advances: platform-driven services/programs and (collectivist) technologies to enhance scheduling, surface resources and support financial well-being.

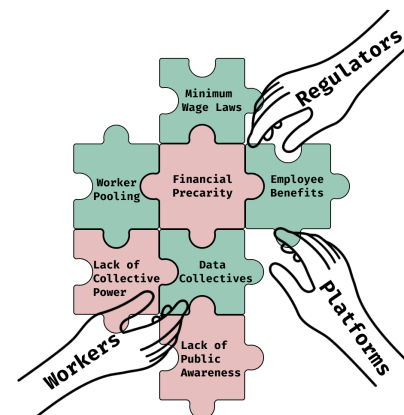


Fig. 2 Multi-Stakeholder Co-design of Solutions

Uncovering collective visions for infrastructural and regulatory change is a critical first step, but to achieve these in practice? In a CSCW '24 workshop, I led like-minded scholars to explore *worker data collectives* [6] for enacting counterdata production as well as advocacy and policy advances. I also interviewed policy experts from varying levels of office to surface prioritized initiatives for data-sharing (*i.e.* safety, fair pay) and design recommendations [5] that subsequently informed early-stage designs of [Gig2Gether](#) (Fig. 3), an open-sourced, cross-platform (pink-, white- and blue-collar domains) web app allowing users to track and share gig work data, view aggregate data visualizations and post qualitative stories about labor conditions that promote evidence-based advocacy and policymaking [6]. In a 7-day field evaluation with 16 participants, **Gig2Gether facilitated cross-platform mutual support** and financial reflection/planning, in addition to revealing latent worker desires for future use cases — including **collective algorithmic auditing and policy advocacy**. For one Gig2Gether user, the tool represented “*a door between [gig workers] and politicians*”.

③ Building Social Solidarity & Pluralistic Interactions amidst Evolving Tech [3, 10]

Alongside technological interventions [2], regulators in our workshops also called for increased and individualized policy. In response, I synthesized legal and policy documents, labor regulations, exemplary state bills and HCI scholarship to envision advances. In an award-winning position paper [3], I advocated for targeted and pluralistic policies that consider diverse task domains and intersectional worker backgrounds. But

while policymakers call for stronger labor regulations, platforms continuously rely on workers' non-employee status to evade employer accountabilities – causing them to shoulder *vulnerabilities like immaterial labor and emotional burdens, alone* [c]. Meanwhile, HCI research has overlooked the role of consumers in worker advocacy, which harms workers when uninformed consumers hold unrealistic expectations of service quality.

How to design socially approachable interventions to **motivate consumer-worker solidarity**? To catalyze *worker-consumer co-liberation* for ridehail, I explored how gamification can motivate passenger understanding and behavioral change around latent driving conditions [10]. Probing 15 passengers and 19 drivers with six interactive in-ride games during workshops, I uncovered passenger knowledge gaps (e.g. long-term health impacts, rating-imposed service expectations) as well as desires for more socially appropriate passenger-driver interactions from both stakeholders, opening opportunities for consumer-led advocacy.

Future Research Aims

I am eager to apply my multi-stakeholder expertise to guide emerging workforces *navigate transformative disruptions of rapidly-advancing innovations* at individual, relational and community-based scales.

Designing Futures that Capture Labor & Evolve Individual Workers. Outside labor relations, workers and consumers increasingly produce value and data for platforms. As AI permeates knowledge and creative work domains, future generations will require new tools and frameworks to augment their unique human capabilities (e.g., intellect [d], creativity, empathy) while preserving humanistic values (e.g., dignity, autonomy, meaning) – when collaborating with AI. In support of creative professionals, I advised an HCI+AI team to design and evaluate POET, which identifies and expands visual homogeneities of text-to-image models to **automatically generate more diverse outputs** [9], and is extensible to modalities beyond images. I am eager to continue exploring techniques that preserve user agency and intent in early-stage pipelines.

In relatively service-based, low-income and vulnerable forms of work, various latent stressors and labor remain undocumented and unacknowledged. As labor increasingly modularizes, emergent sensing capabilities (e.g., video, speech) offer more lightweight pathways for capturing tacit knowledge and latent human expertise. Combining empirical methods from labor studies and organizational science with technical possibilities from engineering or data science (e.g., embedded design, visualization, graphics), I aim to develop richer, more effortless mechanisms to help workers *visibilize, legitimize and revalue labor*.

Reconfiguring Social Relations to Motivate Digital Collective Actions. Gamification [10] represented one method of engaging consumer solidarity with workers through immersion. Another way to promote unity is to collectively recognize that ***gig workers are also consumers*** who benefit from platform resources – e.g. client pools, matching algorithms. I plan to apply frameworks from critical computing and behavioral economics to analyze and design mechanisms/relations that mobilize more bottom-up, consumer-led accountability.

Beyond gig labor markets, labor protections are falling behind accelerating technological growth and evolving worker needs, causing union membership to decline (currently <10% of US adults [e]). I will take a multi-stakeholder perspective to explore novel interventions – leveraging creative and novel techniques such as game or mechanism design – to better align workers and their allies toward scalable peer support.

Building Collaborative, Pluralistic and Trust-based Communities. Online communities of today (e.g. social media, open source development) suffer from misaligned incentives and disruption – e.g. inappropriate comments [4]. Trusting, sustainable communities require shared understandings of on-the-ground realities, negotiations to reconcile pluralistic values. PolicyCraft facilitates deliberation using human and AI-generated cases [8]. Many emergent capabilities (e.g., agent-based simulations, world-building) carry potential to scalably transform governance and our understandings of each other. I intend to investigate more unremarkable [f] systems that promote more prosocial, deliberative and inclusive communities, starting with familiar domains (e.g. gig work, art [12]) before exploring newer realms – e.g. music, moderation, game design [10].